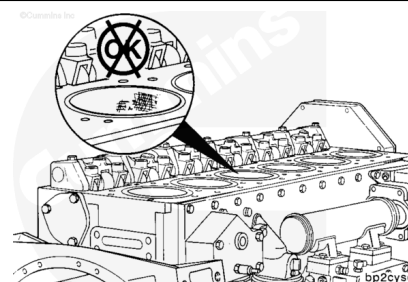


Trainings ▾

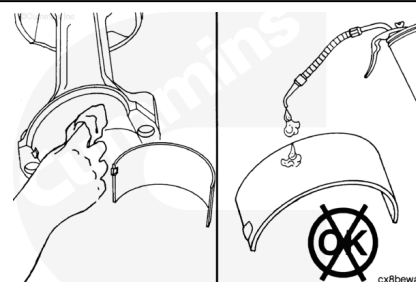
Install

The cylinder block and all parts **must** be clean before assembly. Refer to Procedure 001-028 in Section 1 (</qs3/pubsys2/xml/en/procedures/36/36-001-028.html>) in Troubleshooting and Repair Manual, M11 STC, CELECT and CELECT Plus Engines, Bulletin 3666139. Refer to Procedure 001-028 in Section 1 (</qs3/pubsys2/xml/en/procedures/43/43-001-028.html>) in Troubleshooting and Repair Manual, L10G Natural Gas Base Engine, Bulletin 3666207 for L10G engines to inspect the cylinder liners for reuse.



Use a clean, lint free cloth to clean the connecting rods and bearing shells.

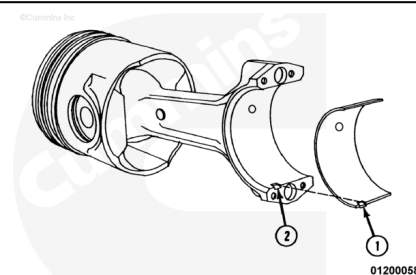
Do **not** lubricate the back side of the bearing shells. The operating clearance of the bearing will be reduced and the bearing can be damaged during engine operation.



If new bearings are **not** used, the used bearings **must** be installed on the same connecting rod from which they were removed.

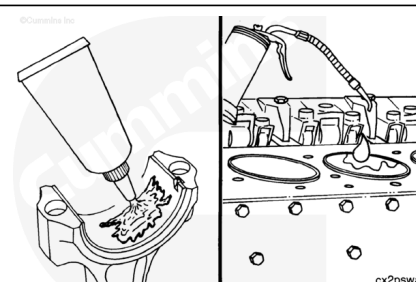
Install the upper bearing shell into the connecting rod.

The tang of the bearing shell (1) **must** be in the slot of the rod (2). The end of the bearing shell **must** be even with the cap mounting surface.



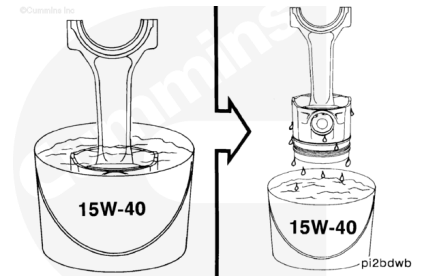
Use Lubriplate™ 105, or equivalent, to coat the inside diameter of the bearing shell.

Apply a heavy film of clean 15W-40 oil to the liner.



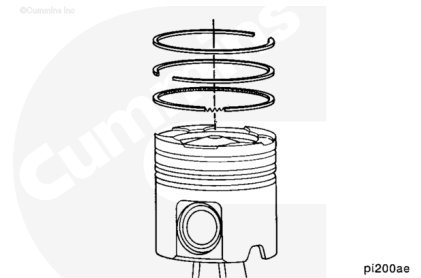
Install the piston and ring assembly in a container of clean 15W-40 oil.

Remove the piston and ring assembly from the container, and let the excess oil drain from the piston.



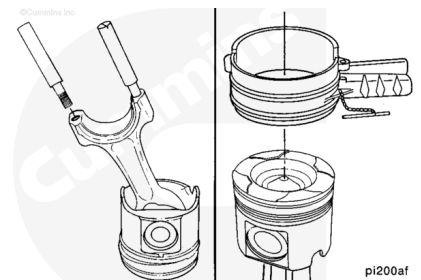
Rotate the rings to position the ring gaps as shown.

Note : The ring gap of each ring **must not** be aligned with the piston pin or with any other ring. If the ring gaps are **not** aligned correctly, the rings will **not** seal properly.

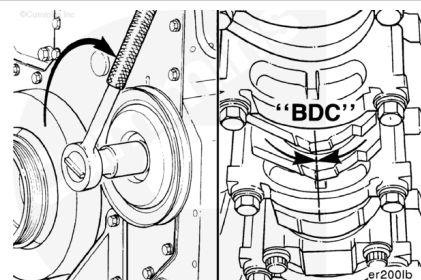


Install connecting rod guide pins, Part Number 3376038.

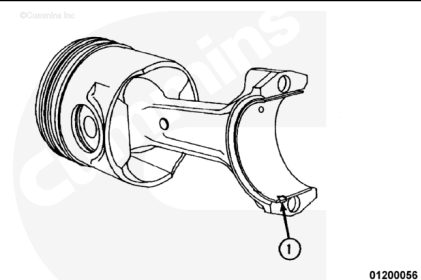
Use piston ring compressor, Part Number 3823309, to compress the rings.



Use the accessory drive pulley to rotate the crankshaft so the connecting rod journal of the connecting rod being installed is at bottom dead center.



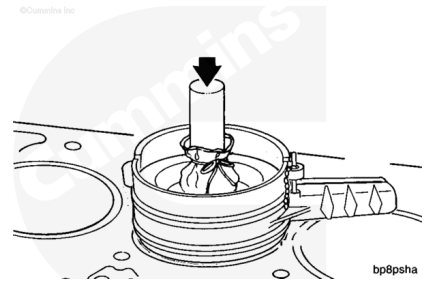
Insert the connecting rod through the cylinder liner with the bearing tang (1) toward the camshaft side of the engine until the ring compressor contacts the top of the liner.



Do not use a metal drift to push the piston into the cylinder liner. The piston rings or cylinder liner can be damaged.

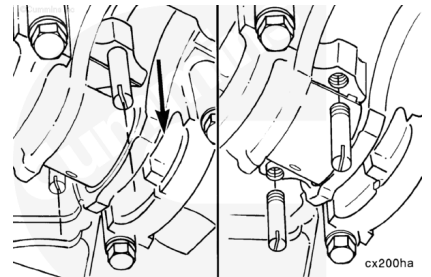
Hold the ring compressor against the cylinder liner. Push the piston through the ring compressor and into the cylinder liner. Push the piston until the top ring is completely in the cylinder liner.

Note : If the piston does not move freely, remove the piston and inspect for broken or damaged rings.



Use the nylon guide pins to align the connecting rod with the crankshaft while pushing the piston and rod assembly in place.

Remove the nylon guide pins.



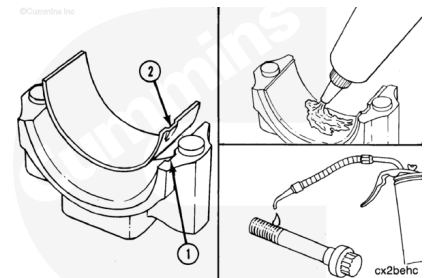
Note : If new bearings are not used, the used bearings **must** be installed on the same connecting rod cap from which they were removed.

Install the bearing in the connecting rod cap.

The tang of the bearing (2) **must** be in the slot of the cap (1).

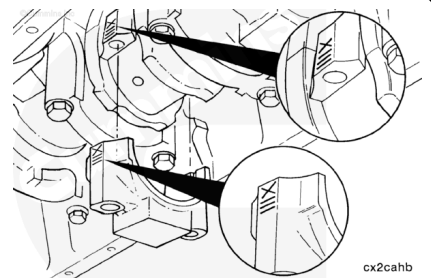
Use Lubriplate™ 105, or equivalent, to coat the inside diameter of the bearing shell.

Use clean 15W-40 oil to lubricate the connecting rod capscrew threads.



The connecting rod and cap **must** have the same number and **must** be installed in the proper cylinder. The connecting rod cap number and rod number **must** be on the same side of the connecting rod to reduce the possibility of engine damage during engine operation.

Install the connecting rod caps and capscrews.

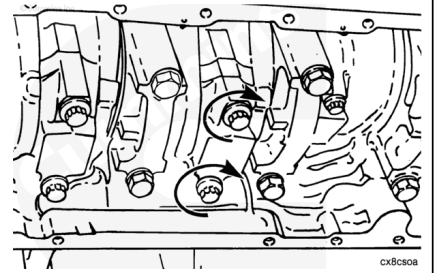


Complete the following steps to tighten the rod capscrews in alternating sequence:

Torque Value:

1. 68 n•m [50 ft-lb]

2. 142 n•m [105 ft-lb]
3. 210 n•m [155 ft-lb]
4. Loosen completely and repeat steps 1 through 3



Measure the connecting rod side clearance.

Connecting Rod Side Clearance		
mm		in
0.10	MIN	0.004
0.30	MAX	0.012

